Linear Algebra First Day Warm-Up
Dr. Osborne 2022

The purpose of this warm-up is to get you to start thinking about the possible situations that can occur with linear systems. You have learned about these types of systems before, in algebra II and pre-calculus, but this class will focus on this material in a manner that you have not fully seen before.

Solve each of the following systems. If there is only one solution, just give the solution. If there are infinitely many solutions, state whether the solution set consists of a line or a plane in three-dimensional space and give a parametric form for the equations of that line or plane. If there are no solutions, explain how the planes associated with the three equations are oriented and why there are no solutions. You should be able to do all of these calculations by hand, but you may use a calculator to check your work.

1. $\begin{aligned} 3x-2y+z&=4\\ 2x-y+3z&=-2\\ 4y+z&=3 \end{aligned}$

2. $\begin{aligned} x+y&=1\\ 2x-3y&=4\\ x+6y&=-1 \end{aligned}$

3. $\begin{aligned} x+2y+z&=-4\\ 3x-3y+2z&=6\\ 5x+y+4z&=-2 \end{aligned}$

4. $\begin{aligned} 3x+2y-z&=1\\ 6x+4y-2z&=2\\ -3x-2y+z&=-1 \end{aligned}$

5. $\begin{aligned} x+y-z&=7\\ 2x-y+3z&=-4\\ 7x+y+3z&=5 \end{aligned}$

6. $\begin{aligned} 5x+7y-3z&=8\\ 3x-5y+2z&=3\\ x+3y-z&=10 \end{aligned}$

In problems 7 – 10, determine the restrictions on *a* required in order for the system to have a single (unique) solution. Can you determine a value of *a* for which the system has no solutions? Infinitely many?

7. $\begin{aligned} 3x-2y&=7\\ ax+3y&=9 \end{aligned}$

8. $\begin{aligned} 2x+y&=a\\ 3x-ay&=4 \end{aligned}$

9. $\begin{aligned} 2x+y&=a\\ -x+4y&=-6\\ 3x-2y&=8 \end{aligned}$.

10. $\begin{aligned} 2x+ay-4z&=3\\ -x+6y+2z&=1 \end{aligned}$.